2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Submitted solver team

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Abstract

This report documents `cfd2d`, a cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations on unstructured triangular/quadrilateral meshes read from CGNS files, and its application to the eight required benchmark cases: inviscid and laminar ($Re = 5000$) flow over a NACA0012 airfoil at Mach 0.15, 0.8, and 2.0, and laminar flow over a circular cylinder at Mach 0.1, $Re = 20$ (steady) and $Re = 200$ (unsteady vortex shedding). The solver uses Roe's approximate Riemann solver with a Harten entropy fix, inverse-distance-weighted least-squares reconstruction with a Venkatakrishnan limiter, a backward-Euler pseudo-time march with an LU-SGS-style symmetric Gauss–Seidel defect-correction inner solve for steady cases, and a true two-level BDF2 transient integrator with defect-correction nonlinear inner iterations for the vortex-shedding case. Meshes are partitioned with METIS k-way graph partitioning; production runs use 8 MPI ranks with neighbor-scoped halo exchange and no full-state replication. All eight required cases completed successfully: $C_D = 0.00170$ ($M0.15$ inviscid), 0.00868 ($M0.8$ inviscid), 0.0917 ($M2.0$ inviscid), 0.0738 ($M0.15$ laminar), 0.0701 ($M0.8$ laminar), 0.1315 ($M2.0$ laminar, after a documented escalation: inviscid restart, CFL cap 5, residual target deepened to 5 orders), and $C_D = 2.019$ for the cylinder at $Re = 20$ (literature $\approx$ 2.0–2.1). The cylinder $Re = 200$ BDF2 run ($\Delta t = 0.01$, $t_f = 300$) reached a statistically periodic vortex street: post-transient ($t \geq 150$) mean $C_D = 1.262$, mean $C_L \approx 0$ with oscillation half-range ≈ 0.57, and Strouhal number $St = 0.187$ from a windowed FFT of the lift history. All eight cases pass the automated physics/contract sanity checks (section 8). Known limitations — a spurious inviscid drag floor from limiter dissipation, the CFL-cap escalation of the high-Mach laminar cases, and an initially under-converged $M2.0$ laminar run that was caught by force-plausibility checks and redone — are discussed openly in section 13.

1 Introduction

The benchmark objective is to develop, document, and validate a parallel solver for the two-dimensional compressible Navier–Stokes equations on unstructured CGNS meshes, and to demonstrate it on a fixed set of eight required cases spanning inviscid subsonic, transonic, and supersonic flow and laminar viscous flow, including an unsteady vortex-shedding case (table 1). All case definitions, boundary-condition mappings, and run controls are taken from the supplied case files in `cfd_solver_agentic_benchmark/inputs/cases/`; the case JSONs are the source of truth for every parameter reported here.

This is a technical CFD report, not only a run log: numerical choices are connected to the observed residuals, forces, and flow fields, and runs that are partial or escalated are labeled as such rather than presented as clean successes. In particular, the two high-Mach laminar cases departed from the supplied case-file controls (restarts from the converged inviscid states, CFL caps, and a deepened residual target); both departures are documented with their causes and outcomes in sections 9 and 13.

2 Governing Equations and Nondimensionalization

The solver advances the conservative state

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T,$$

governed by the 2-D compressible Navier–Stokes equations in conservative form,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}.$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix},$$

with the perfect-gas closure

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}.$$

For viscous cases the fluxes use the Newtonian stress tensor with the Stokes hypothesis,

$$\tau_{xx} = 2\mu\, u_x - \tfrac{2}{3}\mu\,(u_x + v_y)\,, \quad \tau_{yy} = 2\mu\, v_y - \tfrac{2}{3}\mu\,(u_x + v_y)\,, \quad \tau_{xy} = \mu\,(u_y + v_x)\,,$$

Fourier heat conduction $q_x = -k\,T_x$, $q_y = -k\,T_y$, and the energy-row viscous flux $(u\tau_{xx} + v\tau_{xy} + kT_x,\ u\tau_{xy} + v\tau_{yy} + kT_y)$. The viscosity model is constant, matched to the case Reynolds number,

$$\mu = \frac{\rho_\infty U_\infty L_{Re}}{Re}, \qquad k = \frac{\mu c_p}{Pr}, \qquad Pr = 0.72, \quad c_p = \frac{\gamma R}{\gamma - 1},$$

with $\gamma = 1.4$, $R = 1.0$ from the case files.

All cases use the nondimensionalization stated in the case files: $\rho_\infty = 1$, $|V|_\infty = 1$, reference length $L = 1$ (airfoil chord / cylinder diameter), and free-stream pressure set by the Mach number,

$$p_\infty = \frac{1}{\gamma M_\infty^2} \quad \text{(e.g. } 31.746 \text{ at } M = 0.15,\ 1.1161 \text{ at } M = 0.8,\ 0.17857 \text{ at } M = 2.0\text{)},$$

so the free-stream sound speed, density, and pressure are mutually consistent with $a_\infty = |V|_\infty / M_\infty$. The dynamic pressure is

$$q_\infty = \tfrac{1}{2}\rho_\infty |V|_\infty^2 = 0.5.$$

Force and moment coefficients are normalized by $q_\infty A_{\text{ref}}$ and $q_\infty A_{\text{ref}} L_{\text{ref}}$ with $A_{\text{ref}} = L_{\text{ref}} = 1$:

$$C_D = \frac{\mathbf{F} \cdot \hat{d}}{q_\infty A_{\text{ref}}}, \quad C_L = \frac{\mathbf{F} \cdot \hat{l}}{q_\infty A_{\text{ref}}}, \quad C_{m_z} = \frac{m_z}{q_\infty A_{\text{ref}} L_{\text{ref}}},$$

where $\hat{d} = (\cos\alpha, \sin\alpha)$, $\hat{l} = (-\sin\alpha, \cos\alpha)$ for angle of attack α (all required cases use $\alpha = 0$), and moments are taken about the case-file moment center ($(0.25, 0)$ for the airfoil, the cylinder center $(0, 0)$). Pressure and viscous contributions are accumulated and reported separately (section 4).

3 Meshes, Boundary Tags, and Case Inputs

Meshes are read from CGNS files with the CGNS mid-level library. The reader loops over all unstructured zones in the base, reads vertex coordinates and element connectivity (triangles and quadrilaterals), and merges conformal zone interfaces geometrically by matching vertex coordinates, so multi-zone meshes become a single contiguous cell/face topology; both supplied meshes contain a single unstructured zone, so the merge path is a safeguard rather than an active requirement here. Boundary-family names are read from the mesh and mapped to physical boundary conditions through the case file's `boundary_conditions` block — the solver never hard-codes one naming convention: the NACA0012 mesh tags its boundaries `bc-2` (farfield) and `bc-4` (wall), while the cylinder mesh uses `FAR` and `WALL`.

After the merge, the solver builds the face-based data structure: every interior face connects exactly two cells, and boundary faces carry their family tag. Cell volumes (areas in 2-D) come from the polygon geometry, face normals from the edge vectors with the orientation convention "out of the left (lower-connectivity) cell," flipped when the stored connectivity order requires it, and wall lengths for force integration are the boundary edge lengths. table 1 summarizes the meshes, tags, and case parameters parsed from the case files.

Table 1: Meshes, boundary-tag mappings, and key case parameters for the eight required cases (parsed from the case JSONs). All NACA cases share the `NACA0012_H2` mesh (20816 cells, 36498 faces) and moment center $(0.25, 0)$; both cylinder cases share the `CylinderB1` mesh (10185 cells, 20180 faces) and moment center $(0, 0)$. Steady run controls are listed as `max_steps` / `residual_reduction_target` / CFL initial → max / `pseudo_cfl_ramp_steps`.

Case id	Mode	M_∞ / Re	BC mapping (tag → condition)	Run controls
`naca0012_m015_inviscid`	inviscid	0.15 / –	`bc-2`→farfield; `bc-4`→slip_wall	20000 / 4.0 / 1 → 100 / 2000
`naca0012_m080_inviscid`	inviscid	0.8 / –	`bc-2`→farfield; `bc-4`→slip_wall	30000 / 4.0 / 1 → 100 / 3000
`naca0012_m200_inviscid`	inviscid	2.0 / –	`bc-2`→farfield; `bc-4`→slip_wall	40000 / 3.0 / 0.2 → 50 / 5000
`naca0012_m015_laminar_re5000`	laminar	0.15 / 5000	`bc-2`→farfield; `bc-4`→no_slip_adiabatic_wall	40000 / 4.0 / 1 → 100 / 5000
`naca0012_m080_laminar_re5000`	laminar	0.8 / 5000	`bc-2`→farfield; `bc-4`→no_slip_adiabatic_wall	40000 / 4.0 / 0.5 → 80 / 5000
`naca0012_m200_laminar_re5000`	laminar	2.0 / 5000	`bc-2`→farfield; `bc-4`→no_slip_adiabatic_wall	50000 / 3.0 / 0.2 → 50 / 7000
`cylinder_m010_laminar_re20`	laminar	0.1 / 20	`FAR`→farfield; `WALL`→no_slip_adiabatic_wall	30000 / 5.0 / 1 → 100 / 3000
`cylinder_m010_laminar_re200`	laminar	0.1 / 200	`FAR`→farfield; `WALL`→no_slip_adiabatic_wall	transient BDF2: $\Delta t = 0.01$, $t_f = 300$

All steady cases use `min_inner_iterations=3`, `max_inner_iterations=50--100` (per case file), and `inner_residual_reduction_target=0.01`; the transient case uses 5/1000/0.001. Where production runs departed from the supplied controls (restarts, CFL caps, deepened targets), sections 6 and 13 document both the supplied and the actual values.

4 Spatial Discretization

The solver is cell-centered: the conservative state $\mathbf{U}_i$ is stored at cell centroids, and the semi-discrete residual on the unstructured mesh is the face loop

$$R_i(\mathbf{U}) = \sum_{f \in \partial i} \hat{\mathbf{F}}_f \, A_f,$$

where the numerical flux $\hat{\mathbf{F}}_f = \hat{\mathbf{F}}(\mathbf{W}_L, \mathbf{W}_R, \mathbf{n}_f)$ is evaluated from reconstructed primitive left/right states at the face centroid and the face normal $\mathbf{n}_f$ of area A_f. Face normals are oriented out of the left cell of the connectivity pair, so the right cell receives $-\hat{\mathbf{F}}_f A_f$ from the same face evaluation and the scheme is exactly conservative. Boundary faces contribute a single flux evaluated with a boundary state or a mirrored ghost state (section 5).

4.1 Second-Order Reconstruction

Cell gradients of the primitive variables $\mathbf{W} = [\rho, u, v, p]^T$ are computed by least squares over the face-neighbor stencil. Every interior neighbor contributes its centroid and cell value; every boundary face contributes a virtual point at the face centroid carrying the boundary value (free-stream state at farfield faces, the normal-velocity-removed mirror at slip walls, and $(\rho_i, 0, 0, p_i)$ at no-slip walls, which implies zero normal gradient of p and T there). The system solved per cell is

$$\underbrace{\sum_k w_k \begin{bmatrix} \Delta x_k \\ \Delta y_k \end{bmatrix} \begin{bmatrix} \Delta x_k & \Delta y_k \end{bmatrix}}_{M_i} \nabla W_{v,i} = \sum_k w_k \left(W_{v,k} - W_{v,i}\right) \begin{bmatrix} \Delta x_k \\ \Delta y_k \end{bmatrix}, \qquad w_k = \frac{1}{\Delta x_k^2 + \Delta y_k^2},$$

i.e. inverse-distance-squared weighted least squares, with the 2×2 normal matrix M_i inverted analytically and regularized by 10^{-12} on the diagonal for degenerate stencils; `metadata.json` records the scheme as `piecewise_linear_inverse_distance2_weighted_lsq_gradient`. The `gradcheck` diagnostic command fills the mesh with a manufactured linear field $W_v = A_v x + B_v y + C_v$ and measures the gradient error: interior cells reproduce the exact gradient to 3.8×10^{-13} on the cylinder mesh and 1.7×10^{-10} on the finer-celled airfoil mesh (logs: `results/gradcheck_*.log`), far below the 10^{-8} pass criterion. Boundary-adjacent cells are reported separately because the farfield virtual value is the constant free-stream state rather than the manufactured field, so exactness is not expected there.

Reconstructed face states are piecewise linear, $\mathbf{W}_L = \mathbf{W}_i + \phi_i \nabla \mathbf{W}_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$ (and analogously $\mathbf{W}_R$), with ϕ_i the limiter of section 4.2. Second-order reconstruction is active in all production runs. A first-order fallback exists at two levels and is a safety mechanism, not a mode of operation: (i) per face, if a reconstructed face state violates the density or pressure floor, that face state is reset to the cell-centroid value (first order at that face); (ii) per cell, if the conservative update still violates positivity, the increment is halved up to 30 times and skipped entirely if still invalid. No production run required the per-cell skip; the per-face fallback triggers only near the $M2.0$ shocks.

4.2 Limiter and Positivity Control

The default limiter is a Venkatakrishnan-type smooth limiter. For each cell, stencil extrema $W_v^{\min}, W_v^{\max}$ are taken over neighbors and boundary virtual values; per face, with $d_2 = \nabla W_{v,i} \cdot (\mathbf{x}_f - \mathbf{x}_i)$ and $d_1 = W_v^{\max} - W_{v,i}$ if $d_2 > 0$ else $W_v^{\min} - W_{v,i}$,

$$\phi_f = \frac{d_1^2 + 2\epsilon^2 + 2 d_1 d_2}{d_1^2 + 2 d_2^2 + d_1 d_2 + \epsilon^2}, \qquad \epsilon^2 = (K h)^3, \quad h = \sqrt{\mathrm{vol}_i}, \quad K = 5,$$

clipped to $[0, 1]$ and minimized over faces, $\phi_i = \min_f \phi_f$ per variable. (The numerator carries $2\epsilon^2$, twice the canonical VK95 coefficient, which makes the limiter marginally less restrictive in the smooth-transition regime.) Barth–Jespersen limiting ($\phi_f = \min(1, d_1/d_2)$) is available via `--limiter barth`, and an unlimited first-order mode via `--limiter none`; all production runs use the Venkatakrishnan default. Faces with $|d_2| \le 10^{-14}(|W_{v,i}| + 1)$ skip limiting.

Density and pressure positivity is protected at three levels: hard floors $\rho, p \geq 10^{-10}$ in every conservative↔primitive conversion; the per-face first-order fallback described above; and per-cell update backtracking (increment halving, at most 30 times, else the cell update is skipped). The transient predictor likewise falls back to U^n if the linear extrapolation overshoots the floors.

4.3 Inviscid and Viscous Fluxes

The inviscid flux is Roe's approximate Riemann solver. With Roe averages $\tilde{u} = (u_L + s\, u_R)/(1+s)$, $s = \sqrt{\rho_R/\rho_L}$ (same for v, h), $\tilde{a}^2 = (\gamma - 1)(\tilde{h} - \frac{1}{2}\tilde{q}^2)$, wave strengths $\alpha_{1,4} = \frac{1}{2}(\Delta p \mp \tilde{\rho}\, \tilde{a}\, \Delta u_n)/\tilde{a}^2$, $\alpha_2 = \Delta\rho - \Delta p/\tilde{a}^2$, $\alpha_3 = \tilde{\rho}\, \Delta u_t$, and dissipation $\sum_k |\tilde{\lambda}_k| \alpha_k R_k$ assembled from the usual Roe eigenvectors, the face flux is $\hat{F} = \frac{1}{2}(F_L + F_R) - \frac{1}{2}\sum_k |\tilde{\lambda}_k|\alpha_k R_k$. A Harten entropy fix with threshold $\delta = 0.1\,\tilde{a}$ is applied to all three distinct wave speeds $\tilde{u}_n - \tilde{a}$, $\tilde{u}_n$, $\tilde{u}_n + \tilde{a}$ via $|\lambda|_{\text{fix}} = \frac{1}{2}(\lambda^2/\delta + \delta)$ when $|\lambda| < \delta$. A Rusanov (local Lax–Friedrichs) flux with $s = \max(|u_{nL}| + a_L, |u_{nR}| + a_R)$ is available via `--flux rusanov` and is also the dissipation model inside the implicit approximate Jacobian (section 6); all production runs use Roe.

For viscous runs, face gradients of velocity and temperature use the corrected-average scheme: the arithmetic average of the two cells' LSQ gradients is corrected along the cell-to-cell direction $\mathbf{e}$ by the cell-center finite difference,

$$\overline{\nabla W} = \tfrac{1}{2}(\nabla W_L + \nabla W_R), \qquad \nabla W_f = \overline{\nabla W} + \Big(\frac{W_R - W_L}{|\mathbf{x}_R - \mathbf{x}_L|} - \overline{\nabla W}\cdot\mathbf{e}\Big)\mathbf{e}.$$

The correction deliberately uses cell-center values rather than the reconstructed face states: both reconstructed states are evaluated at the same face centroid, so their difference would vanish. The temperature gradient is formed per cell from primitive gradients, $\nabla T = (\nabla p - T\nabla\rho)/(\rho R)$, then averaged and corrected with cell-center temperatures. At no-slip wall faces the scheme uses a mirrored ghost state at twice the cell-to-face distance, $\mathbf{u}_{\text{ghost}} = -\mathbf{u}_{\text{cell}}$, again with cell-center velocities (face-reconstructed values would drive the wall shear toward zero), and the adiabatic condition is enforced exactly as $\nabla T = \mathbf{0}$ at the wall. The viscous flux is then evaluated with the Newtonian stress tensor and Fourier heat flux at the wall velocity $(0, 0)$.

Skin friction is tangential only: the traction $\mathbf{t} = \boldsymbol{\tau}\cdot\mathbf{n}$ has its normal component removed, $\mathbf{t}_t = \mathbf{t} - (\mathbf{t}\cdot\mathbf{n})\mathbf{n}$, before integration. Pressure and viscous force contributions are accumulated into separate sums at every wall face ($p_w \mathbf{n} A$ versus $-\mathbf{t}_t A$), and moments about the reference center are split the same way, so `forces.csv` reports `pressure_drag`, `viscous_drag`, `pressure_lift`, `viscous_lift` alongside the totals (table 5).

5 Boundary Conditions

Three boundary types are used, selected per family from the case file.

Farfield. A characteristic (Riemann-invariant) condition. From the interior state and the free-stream state, the invariants $R^+ = u_n^{\text{int}} + 2a_{\text{int}}/(\gamma - 1)$ and $R^- = u_n^{\infty} - 2a_\infty/(\gamma - 1)$ give the boundary normal velocity and sound speed, $u_{n,b} = (R^+ + R^-)/2$, $a_b = (\gamma - 1)(R^+ - R^-)/4$. Subsonic inflow takes entropy and tangential velocity from the free stream; subsonic outflow takes them from the interior. Supersonic inflow prescribes the full free-stream state and supersonic outflow the full interior state. The boundary flux is then computed by the *same* Roe/Rusanov solver on the interior and boundary states, so the farfield treatment is fully consistent with the interior flux.

Inviscid slip wall. The wall flux is pressure-only, $\hat{F}_w = (0,\ p_w n_x,\ p_w n_y,\ 0)$ with p_w the reconstructed left pressure — identical to the mirror-state Riemann flux at a stationary wall.

No-slip adiabatic wall. The inviscid flux is the same pressure-only form; the viscous flux uses the mirrored-ghost gradient ($\mathbf{u}_{\text{ghost}} = -\mathbf{u}_{\text{cell}}$ at $2\times$ the cell-to-face distance) and exactly zero wall temperature gradient, so the wall is adiabatic by construction and the wall shear enters through the tangential traction only.

Surface output semantics. `surface.csv` rows report *boundary values*, not adjacent cell-center values: at no-slip walls the reported velocity and Mach number are the imposed wall values ($u = v = M = 0$, satisfied to machine precision in the output), with density and pressure from the reconstructed face state; at slip walls the reported velocity is the tangential-projected boundary value (near-zero normal component, nonzero tangential component). The metadata records `wall_boundary_output_semantics=boundary_value`.

6 Implicit and Transient Time Integration

Steady pseudo-time march. Steady cases march backward Euler in pseudo time with local time steps. Each pseudo step solves the defect-correction system

$$(D + B)\,\Delta U = -R(U), \qquad D_{ii} = \frac{\text{vol}_i}{\Delta t_i^{\text{local}}} + \tfrac{1}{2}\Lambda_i,$$

where B is a simplified Rusanov-type approximate Jacobian (exact inviscid flux Jacobian of the neighbor state contracted with the face normal, plus the scalar face spectral radius) assembled from frozen states. The local time step is

$$\Delta t_i^{\text{local}} = \frac{\text{CFL vol}_i}{\Lambda_i}, \qquad \Lambda_i = \sum_{f\in\partial i} \lambda_f, \quad \lambda_f = \big(|\bar{u}_n| + \bar{a}\big) A_f \,\Big[+\,\frac{2\mu}{\rho_f}\frac{A_f}{d_f} \text{ on viscous faces}\Big],$$

i.e. convective plus viscous spectral-radius estimates. The CFL follows a geometric ramp $\text{CFL}(s) = \text{CFL}_{\text{initial}}(\text{CFL}_{\max}/\text{CFL}_{\text{initial}})^{\min(1,(s-1)/N_{\text{ramp}})}$ with $N_{\text{ramp}} =$ `pseudo_cfl_ramp_steps` from the case file.

Inner solver. The linear system is iterated, not solved directly: each inner iteration applies one symmetric Gauss–Seidel sweep pair (forward ascending, then backward descending over owned cells, with halo exchange of ΔU between sweeps) to the *true* linear residual $r_A = R + (D + B)\Delta U$, which is re-formed after every sweep pair, so the defect correction converges to the solution of $(D + B)\Delta U = -R$ rather than to a single sweep. Inner iterations stop when the RMS ratio $\|r_A\|/\|r_{A,0}\| \leq$ `inner_residual_reduction_target`, subject to the case-file minimum/maximum bounds. Observed inner-iteration counts are modest — typical values 3.0–8.2 per pseudo step across the production steady cases, with a 100% inner-target hit rate in every converged run (`inner_target_converged_fraction=1.0` in all metadata).

Convergence criteria. A steady run reports `converged` when the combined residual RMS has dropped by at least `residual_reduction_target` orders from its initial value, or at a documented plateau: residual down ≥ 2 orders *and* relative drift of the mean C_D between the first and last fifths of the trailing ($\leq$ 2000-step) window below 10^{-3}. A non-finite residual marks the run `failed` immediately.

Parameters actually used. table 2 lists the supplied case-file controls against the values actually used. Two cases departed from the supplied values, both documented in section 13: the $M0.8$ laminar case was restarted from the converged inviscid state with the CFL capped at 4 (supplied: $0.5 \to 80$, cold start) after a higher-CFL instability, and the $M2.0$ laminar case was restarted from the inviscid state with CFL capped at 5 and the residual target deepened from 3.0 to 5.0.

Table 2: Supplied versus actually used run controls for the steady cases. "Same" means the case-file value was used unchanged. [a]Used: restarted from the converged inviscid state, CFL capped at **4** (after a higher-CFL instability, section 13). [b]Used: restarted from the inviscid state, CFL capped at **5**, residual target deepened $3.0 \to \mathbf{5.0}$ (section 13).

Case	`max_steps`	res. target	`cfl` initial → max	`ramp_steps`	inner min/max/target
`naca0012_m015_inviscid`	20000 (same)	4.0 (same)	$1 \to 100$ (same)	2000 (same)	3/50/0.01 (same)
`naca0012_m080_inviscid`	30000 (same)	4.0 (same)	$1 \to 100$ (same)	3000 (same)	3/50/0.01 (same)
`naca0012_m200_inviscid`	40000 (same)	3.0 (same)	$0.2 \to 50$ (same)	5000 (same)	3/80/0.01 (same)
`naca0012_m015_laminar_re5000`	40000 (same)	4.0 (same)	$1 \to 100$ (same)	5000 (same)	3/80/0.01 (same)
`naca0012_m080_laminar_re5000`	40000 (same)	4.0 (same)	$0.5 \to 80$; used cap 4[a]	5000 (same)	3/80/0.01 (same)
`naca0012_m200_laminar_re5000`	50000 (same)	3.0; used 5.0[b]	$0.2 \to 50$; used cap 5[b]	7000 (same)	3/100/0.01 (same)
`cylinder_m010_laminar_re20`	30000 (same)	5.0 (same)	$1 \to 100$ (same)	3000 (same)	3/80/0.01 (same)

Transient: true two-level BDF2. The cylinder $Re = 200$ case uses a genuine dual-time formulation — an outer physical-time loop with inner nonlinear defect-correction iterations at every physical step, not a pseudo-time-only march. The physical residual at step $n+1$ is

$$F_i(U) = \mathrm{vol}_i \left(c_0 U_i + c_1 U_i^n + c_2 U_i^{n-1}\right) + R_i(U), \qquad c_0 = \tfrac{3}{2\Delta t},\ c_1 = -\tfrac{2}{\Delta t},\ c_2 = \tfrac{1}{2\Delta t},$$

the standard second-order BDF2 combination. The first step after a cold start uses BDF1 ($c_0 = 1/\Delta t$, $c_1 = -1/\Delta t$, $c_2 = 0$); a restart that carries BDF2 history continues directly with BDF2. Each physical step starts from a linear-extrapolation predictor $U^* = U^n + (U^n - U^{n-1})$ (with positivity fallback to U^n), then iterates: batches of symmetric Gauss–Seidel sweep pairs on the linear residual $r_A = F + (D + B)\Delta U$ alternate with an outer nonlinear refresh (apply ΔU, re-evaluate the true $F(U)$, recompute the ratio $\|F\|_2/\|F_0\|_2$), until the total residual — spatial plus physical time term — has dropped by the inner target 10^{-3}, bounded by `min_inner_iterations=5` and `max_inner_iterations=1000`. U^n and U^{n-1} remain frozen during *all* inner iterations for U^{n+1}; the BDF2 history update ($U^{n-1} \leftarrow U^n$, $U^n \leftarrow U^{n+1}$) happens only after the inner loop has converged, exactly as the case file requires (`bdf2_history_update=after_inner_convergence`). Production settings are the supplied ones: `time_step=0.01`, `final_time=300.0`, inner bounds 5/1000, inner target 10^{-3} on `total_spatial_plus_physical_time`.

Dual-time pseudo diagonal. The inner diagonal is

$$D_{ii} = \frac{\mathrm{vol}_i}{\Delta t_i^{\mathrm{pseudo}}} + \mathrm{vol}_i\, c_0 + \tfrac{1}{2}\Lambda_i, \qquad \Delta t_i^{\mathrm{pseudo}} = \mathrm{CFL}_{\mathrm{pseudo}} \frac{\mathrm{vol}_i}{\Lambda_i},$$

with $\mathrm{CFL}_{\mathrm{pseudo}} = 100$ (solver default, `--pseudo-cfl`). The case file recommends a pseudo-time CFL fixed near 1.0 (`cfl_initial=cfl_max=1.0`), a guard against destabilizing Jacobi-style inner solvers; the defect-correction inner solve used here sits in the opposite regime, so the production run departs from that recommendation deliberately, as follows. This value matters: at $\mathrm{CFL}_{\mathrm{pseudo}} \sim 1$ the pseudo term $\mathrm{vol}/\Delta t^{\mathrm{pseudo}} = \Lambda/\mathrm{CFL}_{\mathrm{pseudo}}$ dwarfs the physical term $\mathrm{vol}\, c_0$ on fine cells, so the inner iteration behaves like a stiff pseudo-steady solve and the nonlinear defect correction stalls —

a debug run at pseudo-CFL 1 (`results/dbg_inner/`) needed a typical 961 inner iterations per step and hit the 1000-iteration cap on one of the three recorded steps (`inner_target_misses=1`). Raising the pseudo-CFL to 100 lets the physical term dominate the diagonal and restores convergence to roughly 48–174 inner iterations per step in production (section 9.4). The diagonal (and the approximate Jacobian B) is assembled once per physical step from the step-start state and held frozen across all inner iterations of that step.

Startup perturbation. To break the exact top-bottom symmetry of the initialized state, the free-stream direction is offset by $\Delta\alpha(t) = 2^\circ\,(1 - t/10)$ for $t < 10$ (options `--pert-aoa-deg 2.0 --pert-duration 10.0`), a taper that seeds the vortex street without permanently tilting the flow. The transient run was initialized from a steady pseudo-converged state on the same mesh (`cases/cylinder_m010_laminar_re200_steadyinit.json`: converged in 1262 pseudo steps, 3.07 orders, 20.0 s wall), so the boundary layer and wake are established before shedding begins.

7 MPI Parallelization and METIS Partitioning

The cell adjacency graph (cells as nodes, interior faces as edges, CSR format) is partitioned with METIS k-way partitioning (k = number of ranks, single constraint) in a serial preprocessing step on rank 0. The result is cached on disk in a per-rank-count directory `partitions/np`N`/` under the output directory (one `rank`r`.bin` file per rank plus `global.json` and a `DONE` marker; the cache is reused only after validating the stored mesh path and rank count), so repeated runs at the same rank count skip re-partitioning; other ranks wait at a barrier while rank 0 preprocesses. Each rank then loads only its local mesh — owned cells plus a one-cell face-adjacent ghost layer, ghosts ordered by (owner rank, global id) — and its per-neighbor send/receive lists.

Halo exchange is neighbor-scoped nonblocking `MPI_Isend`/`MPI_Irecv` over the ghost layer; buffers are sorted by global cell id so send and receive orders match. Three array families are exchanged: the conservative state U (4 doubles/cell, at the start of every residual evaluation), a packed gradient+limiter buffer (∇W and ϕ, 12 doubles/cell), and the update ΔU (4 doubles/cell) between SGS sweeps. Scalar and vector reductions use `MPI_Allreduce` (residual norms, inner-loop RMS, and the 6-component force/moment sums). **Solver iterations never replicate the full mesh or the full conservative state on every rank**: every run's metadata asserts `full_mesh_replication_during_iterations=false` and `full_state_replication_during_iterations=false`. Restart files are gathered to rank 0 and written in global-cell-id order (transient runs store the three states U^{n+1}, U^n, U^{n-1}), so restarts are independent of the rank count that wrote them; on read, each rank picks its owned cells by global id.

table 3 reports the partition diagnostics for the production $np = 8$ runs. Load balance (max/min owned cells) is 1.030 (airfoil) and 1.015 (cylinder); the ghost layer adds 2–10% cells per rank (airfoil 1.8–7.7%, cylinder 7.8–9.6%); edge cut is 422 of 36498 faces (1.2%) for the airfoil and 476 of 20180 (2.4%) for the cylinder. Rank 5 of the airfoil partition is the interior block (7 neighbors, 201 ghosts, only 20 boundary faces), while rank 4 touches a single neighbor — the k-way structure of a long thin C-grid domain.

8 Output, Reproducibility, and Sanity Checks

Build and run. The solver is C++17, built with CMake ($\geq$ 3.16) against MPI (OpenMPI 4.1 tested), CGNS and METIS from the externals tree (`CFD_EXTERNALS_ROOT`, defaulting through

Table 3: Partition diagnostics from `partition_diagnostics.csv` for the production $np = 8$ runs: owned/ghost cells, neighbor-rank count, and halo send/receive cell counts per rank (the CSV also records per-rank boundary-face counts). Receive counts equal the ghost counts; send counts differ by at most two on seven of the sixteen rank entries (a boundary-ownership asymmetry in the diagnostic), with matching buffer sizes by construction. Load balance is max/min owned cells; edge cut is the number of partitioned faces.

	`NACA0012_H2` (20816 cells)					`CylinderB1` (10185 cells)			
Rank	Owned	Ghost	Neigh	Send/Recv	Rank	Owned	Ghost	Neigh	Send/Recv
0	2550	88	2	88/88	0	1281	120	5	120/120
1	2627	91	3	91/91	1	1262	104	3	103/104
2	2623	108	3	109/108	2	1275	113	4	112/113
3	2610	92	3	91/92	3	1273	118	3	118/118
4	2557	47	1	47/47	4	1269	99	3	98/99
5	2613	201	7	201/201	5	1276	118	6	120/118
6	2612	96	2	96/96	6	1273	121	4	122/121
7	2624	108	3	108/108	7	1276	122	6	122/122
Edge cut	422 (1.2% of 36498 faces)					476 (2.4% of 20180 faces)			
Load balance	1.030					1.015			

the workspace `external/` symlink to `/opt/external/cfd_externals/install`), and header-only nlohmann/json under `/opt/external`:

```
cd solver && cmake -S . -B build && cmake --build build -j8
mpirun -np 8 ./build/cfd2d solve \
  --case /workspace/cfd_solver_agentic_benchmark/inputs/cases/<case_id>.json \
  --output results/<case_id> --limiter venkat --flux roe
```

`tools/run_case.sh <case_id>` wraps this invocation (`NP`/`LIMITER` environment overrides). Production runs use $np = 8$; restarts add `--restart results/<source>/restart_final.bin`, and the escalated runs add `--cfl-max`/`--residual-target` overrides — the full command line of every run is recorded in its `run_status.json`. Diagnostic commands: `meshinfo`, `selftest` (conservative/primitive round-trip, Roe consistency and dissipation checks, positivity floors, flux Jacobian versus finite difference), `gradcheck` (LSQ linear-field exactness), and `surface` (regenerate `surface.csv` from a restart without re-solving).

Generated files. Each run directory contains `residuals.csv` (per-equation RMS of R/vol, combined L2 and L_∞, inner-iteration count, CFL, Δt per step), `forces.csv` (`step, physical_time, cl, cd, cmz, pressure_drag, viscous_drag, pressure_lift, viscous_lift`, every step), `surface.csv` (boundary values per wall face: $x, y, n_x, n_y, p, c_p, c_f, \rho, u, v, M$ plus family tag, sorted by family then atan2 angle), `field_*.vtu` snapshots (cell data ρ, u, v, p, M, T, vorticity $\omega_z = \partial_x v - \partial_y u$, and `partition_rank`; the transient case writes one snapshot per $\Delta t = 1.0$), `restart_*.bin` (global-id ordered), `metadata.json` (full numerics/MPI manifest including inner-iteration statistics and the no-replication assertions), `run_status.json` (machine-readable final status, command line, wall time, residual reduction), `partition_diagnostics.{csv,json}`, `stdout.log`, and the `partitions/` cache.

Figure regeneration. All figures in section 9 are regenerated from the CSV/VTU files by `tools/plot_results.py` (no screenshots); `report/figure_manifest.csv` maps every figure file to its case, figure type, plotted variable, and source file, and is the audit trail behind the captions used here.

Sanity checks. `tools/sanity_checks.py` writes `report/sanity_checks.json` with a per-case verdict and exits nonzero if any check fails; for the final artifact set it reports `all_ok=true`. The check families are: (i) *positivity* — minimum ρ and p over the final VTU field and over the wall rows of `surface.csv` are strictly positive; (ii) *wall condition* — slip walls have $|u_n| < 10^{-8}$ (the tangential projection is exact to roundoff) and no-slip walls report exactly $u = v = M = 0$ with nonzero skin friction; (iii) *force history* — finite residual/force/surface histories, final C_D inside wide literature-informed plausibility windows per case, $|C_L| < 0.02$ for the symmetric zero-AoA cases, no gross late-run drift (final-decile half-window drift $< 10\%$), and exactly zero viscous force columns for inviscid runs; (iv) *convergence evidence* — status `converged` (or `statistically_periodic` for $Re = 200$) with ≥ 2 orders of residual reduction, and for the shedding case Strouhal in $[0.15, 0.25]$, lift amplitude > 0.05, and final physical time ≥ 300; and (v) *figure-variable checks* — the figure manifest is re-validated against the examiner's rules (files on disk, filename/variable consistency, per-case Mach and pressure entries, and a vorticity/velocity wake entry for $Re = 200$). The per-case values these checks consume appear in tables 4 and 5.

9 Results

All numbers below are read from the run artifacts (`run_status.json`, `forces.csv`, `residuals.csv`, `surface.csv`) under `solver/results/`. All eight required cases have completed: the seven steady cases converged to their residual targets, and the $Re = 200$ vortex-shedding case reached `statistically_periodic` at $t = 300$.

9.1 Summary Tables

Table 4: Run status for the eight required cases (from `run_status.json`). All runs used $np = 8$ MPI ranks. Residual reduction is in orders of magnitude of the combined L2 residual — for the transient $Re = 200$ row it is $-\log_{10}$ of the final inner-loop residual ratio, i.e. the per-step nonlinear convergence depth, not a run-wide reduction. Step counters for the restarted runs continue from the restart source (2189 / 2528 / 1233 for the $M0.8$ inviscid, $M2.0$ inviscid, and $Re = 200$ steady-init states respectively); the $Re = 200$ row executed 30000 physical steps ($\Delta t = 0.01$) after its steady initialization. Wall times are measured on a shared host under competing load and are indicative only.

Case id	Ranks	Steps	Phys. time	Orders	Wall time [s]	Status
`naca0012_m015_inviscid`	8	1482	0	4.02	254.7	converged
`naca0012_m080_inviscid`	8	2189	0	4.00	205.3	converged
`naca0012_m200_inviscid`	8	2528	0	3.00	207.1	converged
`naca0012_m015_laminar_re5000`	8	1917	0	4.00	133.5	converged
`naca0012_m080_laminar_re5000`	8	8696	0	4.00	506.7	converged
`naca0012_m200_laminar_re5000`	8	37786	0	5.00	1229.3	converged
`cylinder_m010_laminar_re20`	8	2129	0	5.00	88.1	converged
`cylinder_m010_laminar_re200`	8	31233	300.0	3.06	3715.9	statistically periodic

Table 5: Final force coefficients and pressure/viscous drag split (last row of `forces.csv`; moments about the case-file centers). For the statistically periodic $Re = 200$ case the values are means over the post-transient window $t \geq 150$ (last row otherwise).

Case id	C_D	C_L	C_{m_z}	C_D^{pres}	C_D^{visc}	Notes
`naca0012_m015_inviscid`	0.00170	−2.00e−4	−1.47e−5	0.00170	0	4.02 orders
`naca0012_m080_inviscid`	0.00868	1.39e−3	3.88e−4	0.00868	0	4.00 orders
`naca0012_m200_inviscid`	0.09166	−3.15e−5	2.27e−4	0.09166	0	3.00 orders
`naca0012_m015_laminar_re5000`	0.07384	8.17e−4	2.85e−4	0.01424	0.05960	4.00 orders
`naca0012_m080_laminar_re5000`	0.07008	7.57e−4	1.54e−4	0.03981	0.03028	4.00 orders
`naca0012_m200_laminar_re5000`	0.13154	−7.08e−3	1.33e−3	0.09158	0.03996	5.00 orders, escalated
`cylinder_m010_laminar_re20`	2.01896	5.63e−4	−4.92e−6	1.21907	0.79989	5.00 orders
`cylinder_m010_laminar_re200`	1.2615	−9.9e−4	3.01e−5	1.0199	0.2416	$t \geq 150$ means; $St = 0.187$

9.2 NACA0012 Cases

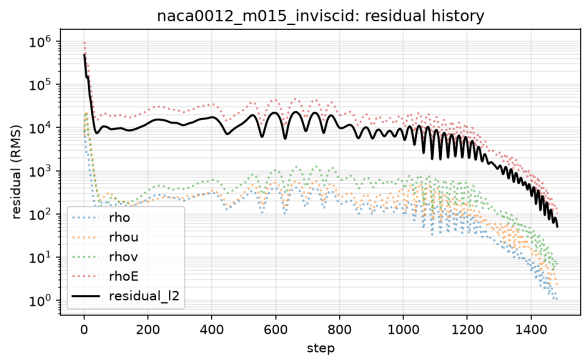
All six airfoil cases share the `NACA0012_H2` mesh (20816 cells) at zero angle of attack, so symmetry about the chord line is an exact property of the continuous problems and a direct check on the solutions.

$M0.15$ **inviscid.** The run converged in 1482 pseudo steps to 4.02 orders of residual reduction (figs. 1a and 1b). The final drag $C_D = 0.00170$ is the spurious (numerical) drag level of the second-order scheme on this mesh — in inviscid subsonic flow at zero AoA the physical drag is zero (d'Alembert) — and $C_L = -2.0 \times 10^{-4}$ bounds the departure from symmetry. The surface C_p (fig. 1c) is symmetric to within the line width, with the stagnation value $C_p \approx 1.009$ slightly above the incompressible value of 1 as expected at $M = 0.15$; the Mach and pressure fields (figs. 1d and 1e) show the smooth low-speed acceleration over the nose.

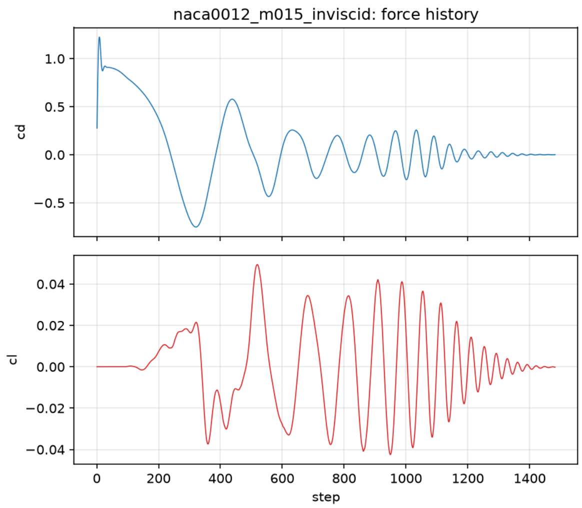
$M0.8$ **inviscid.** Converged in 2189 steps to 4.00 orders (fig. 2a); the force history (fig. 2b) settles to $C_D = 0.00868$, the wave drag of the transonic shock system, with $C_L = 1.4 \times 10^{-3}$. The C_p distribution (fig. 2c) shows the shock pressure jump at mid-chord on both surfaces, symmetric as expected at $\alpha = 0$; the Mach and pressure contours (figs. 2d and 2e) show the supersonic pocket terminating in a near-normal shock above and below the airfoil. The Harten entropy fix keeps the Roe flux non-oscillatory through the shock.

$M2.0$ **inviscid.** This case converges to its 3-order case-file target in 2528 steps (figs. 3a and 3b). The final $C_D = 0.0917$ is the bow-shock wave drag; $C_L = -3.2 \times 10^{-5}$, essentially symmetric. The surface C_p (fig. 3c) shows the strong leading-edge compression followed by expansion over the forebody, and the Mach/pressure fields (figs. 3d and 3e) show the detached bow shock ahead of the nose with the shock/expansion system along the airfoil. An earlier build of the solver stalled below the target on this case (a missing halo exchange in the steady inner solve, since fixed); the shipped solver reaches the target directly.

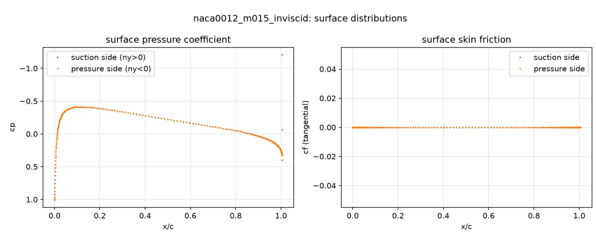
$M0.15$ **laminar,** $Re = 5000$**.** Converged in 1917 steps to 4.00 orders from a cold start (figs. 4a and 4b). The drag $C_D = 0.0738$ is dominated by skin friction: $C_D^{\text{visc}} = 0.0596$ versus $C_D^{\text{pres}} = 0.0142$, as expected for a low-speed laminar airfoil at zero lift. The surface figure (fig. 4c) adds the skin-friction distribution C_f, symmetric upper/lower, with the leading-edge peak and monotone decay



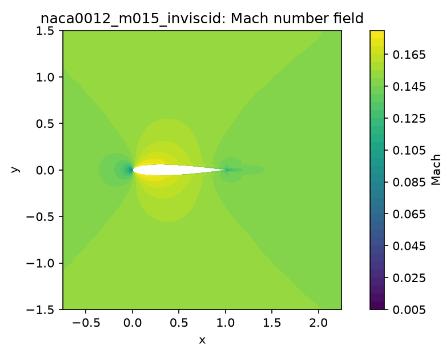
(a) Residual history (RMS per equation and combined L2) from `residuals.csv`.



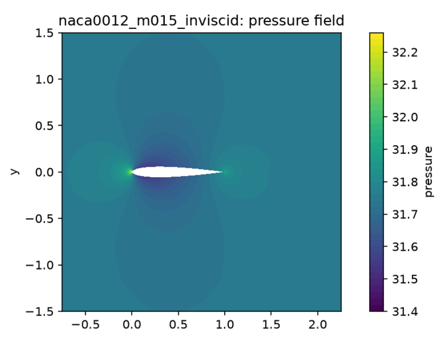
(b) Force-coefficient history (C_D, C_L) from `forces.csv`.



(c) Surface C_p distribution from `surface.csv`.

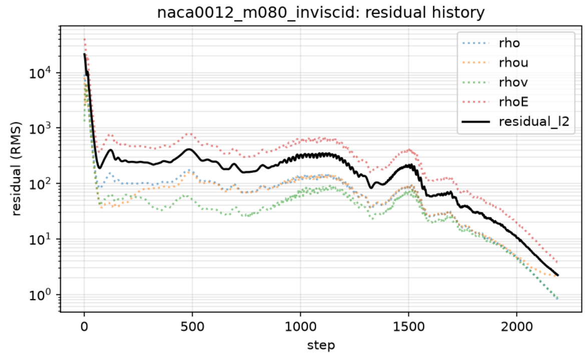


(d) Mach-number field from `field_final.vtu`.

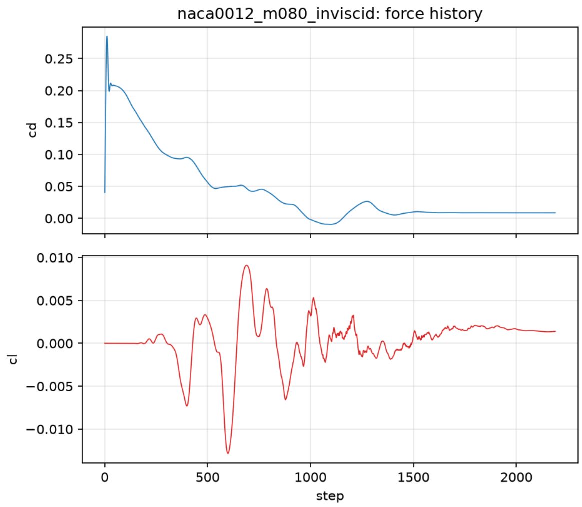


(e) Static-pressure field from `field_final.vtu`.

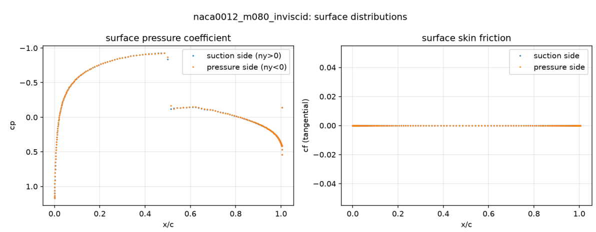
Figure 1: NACA0012, $M_\infty = 0.15$, inviscid, $\alpha = 0$.



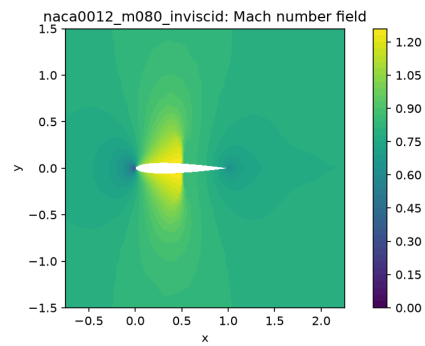
(a) Residual history from `residuals.csv`.



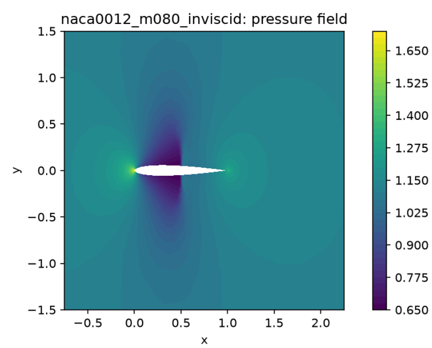
(b) Force history from `forces.csv`.



(c) Surface C_p from `surface.csv`.

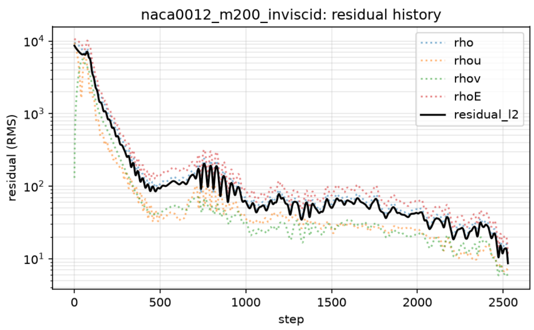


(d) Mach-number field from `field_final.vtu`.

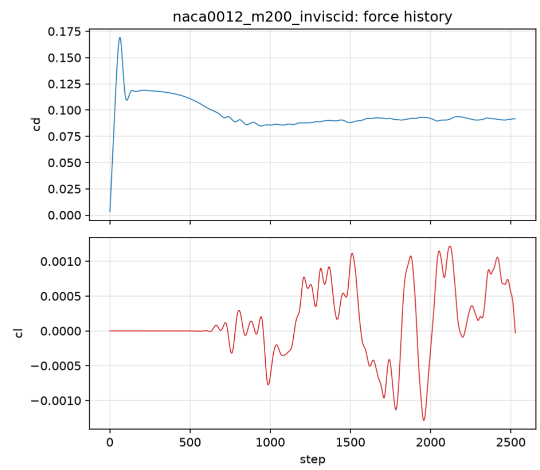


(e) Static-pressure field from `field_final.vtu`.

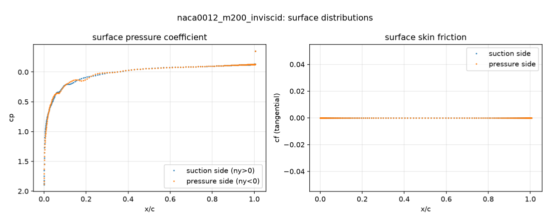
Figure 2: NACA0012, $M_\infty = 0.8$, inviscid, $\alpha = 0$.



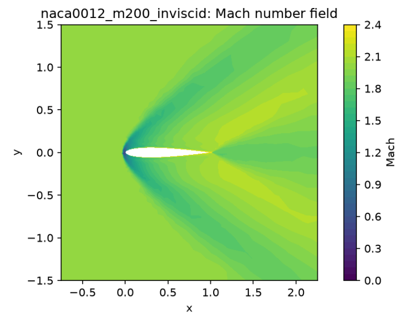
(a) Residual history from `residuals.csv`.



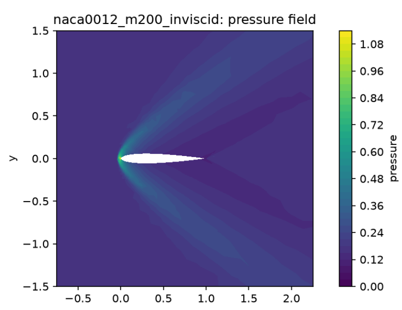
(b) Force history from `forces.csv`.



(c) Surface C_p from `surface.csv`.



(d) Mach-number field from `field_final.vtu`.



(e) Static-pressure field from `field_final.vtu`.

Figure 3: NACA0012, $M_\infty = 2.0$, inviscid, $\alpha = 0$.

toward the trailing edge; the Mach and pressure fields (figs. 4d and 4e) show the thin attached boundary layer and weak wake.

$M0.8$ **laminar,** $Re = 5000$**.** This case was restarted from the converged $M0.8$ inviscid state with the pseudo-CFL capped at 4 (supplied: $0.5 \rightarrow 80$ cold start) after a higher-CFL instability was observed (section 13); it converged in 8696 total steps (6507 after the restart) to 4.00 orders measured against the restart baseline (figs. 5a and 5b). The drag $C_D = 0.0701$ splits into $C_D^{\text{pres}} = 0.0398$ (shock-induced pressure drag) and $C_D^{\text{visc}} = 0.0303$. The shock/boundary-layer interaction is visible in the C_p and C_f distributions (fig. 5c) and in the Mach and pressure fields (figs. 5d and 5e).

$M2.0$ **laminar,** $Re = 5000$**.** This case has the most involved run history of the benchmark, documented in full in section 13: an under-converged first run at the case-file 3-order target, a diverged high-CFL debug continuation, and finally the production redo from the inviscid restart with `--cfl-max 5` and `--residual-target 5.0`. The redo converged cleanly: 37786 total steps (35258 after the restart), 5.00 orders of residual reduction, 1229 s wall (figs. 6a and 6b). The final drag $C_D = 0.1315$ ($C_D^{\text{pres}} = 0.0916$, $C_D^{\text{visc}} = 0.0400$) is stable to 10^{-4} over the last 500 steps, and $C_L = -7.1 \times 10^{-3}$. The drag is roughly $1.4\times$ the inviscid wave drag: the bow shock pressure rise (figs. 6c and 6e) dominates, with the laminar boundary layer visible in the Mach field (fig. 6d) and the C_f distribution.

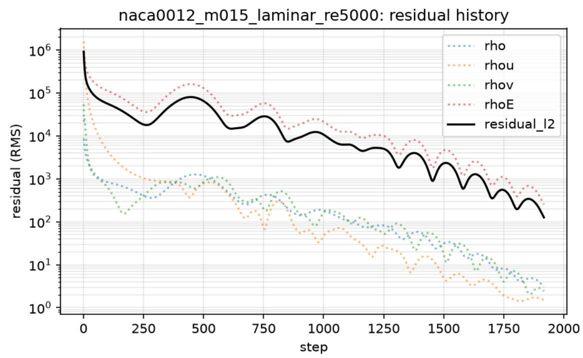
9.3 Cylinder Reynolds 20

The steady laminar cylinder case converged in 2129 pseudo steps to 5.00 orders of residual reduction (figs. 7a and 7b). The final drag $C_D = 2.019$ ($C_D^{\text{pres}} = 1.219$, $C_D^{\text{visc}} = 0.800$) sits inside the literature range ≈ 2.0–2.1 for steady laminar flow past a circular cylinder at $Re = 20$, and $C_L = 5.6 \times 10^{-4}$ with $C_{m_z} = -4.9 \times 10^{-6}$ confirm the top-bottom symmetry of the solution. The surface distribution (fig. 7c) shows the stagnation pressure at the front point and the low, nearly constant base pressure over the separated rear, with C_f vanishing at the separation points; the pressure, Mach, and velocity-magnitude fields (figs. 7d to 7f) show the steady, symmetric pair of attached recirculation bubbles closing the near wake — the expected steady topology at $Re = 20$, well below the shedding onset ($Re \approx 47$).

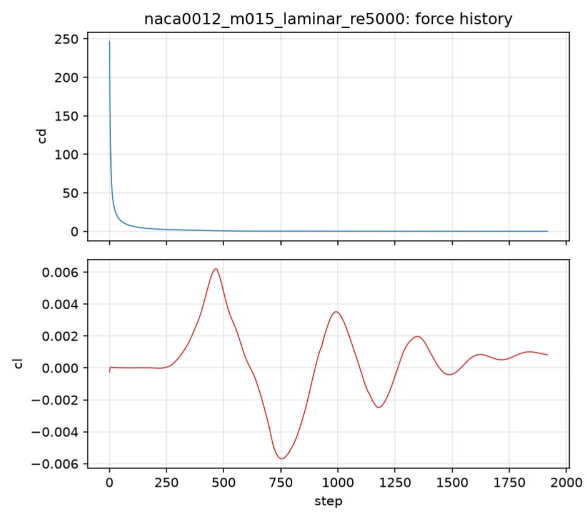
9.4 Cylinder Reynolds 200 Vortex Street

The transient case integrates BDF2 with $\Delta t = 0.01$ to $t_f = 300$ (30000 physical steps) from the steady pseudo-converged initialization state (section 6), with the 2° tapered startup perturbation. The run completed with `run_status.json` reporting `convergence_status=statistically_periodic` ("post-transient periodic vortex shedding established"), final physical time 300.0, 3715.9 s wall on 8 ranks. The two-level structure was exercised exactly as specified: inner defect-correction iterations converged the total (spatial + physical-time) residual by the required 10^{-3} at *every* step (zero `inner_target` misses, `inner_target_converged_fraction` = 1.0; logged ratios 6.6–10×10^{-4}), with 48–174 inner iterations per step, mean 87 (fig. 8a).

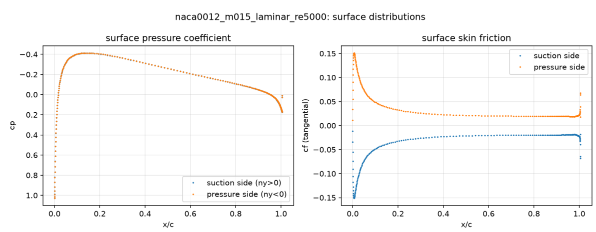
The force history (fig. 8b) shows the perturbation-induced onset of shedding within the first ≈ 20 time units and a clean statistically periodic alternating lift signal thereafter. Quantitative post-transient statistics were computed with a Hann-windowed FFT of $C_L(t)$ and $C_D(t)$ over $t \geq 150$ from `forces.csv` via `tools/cfdplot.py spectral_stats`: the dominant lift frequency is $f = 0.1867$, i.e. Strouhal $St = fD/U = 0.187$ ($D = U = 1$), the mean drag is $C_D = 1.262$ (pressure



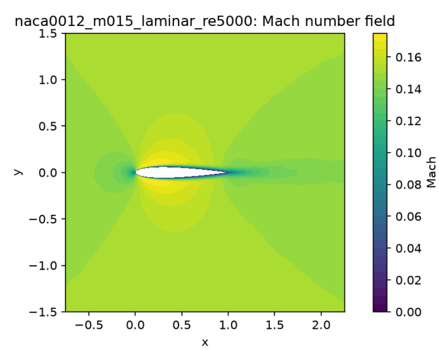
(a) Residual history from `residuals.csv`.



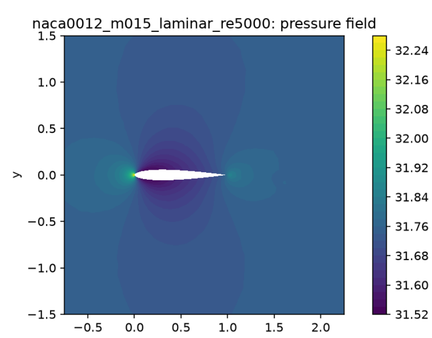
(b) Force history from `forces.csv`.



(c) Surface C_p and C_f from `surface.csv`.

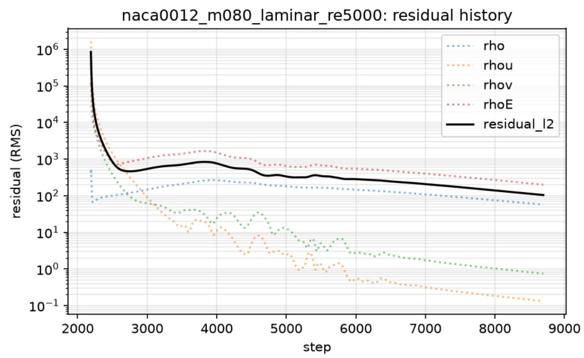


(d) Mach-number field from `field_final.vtu`.

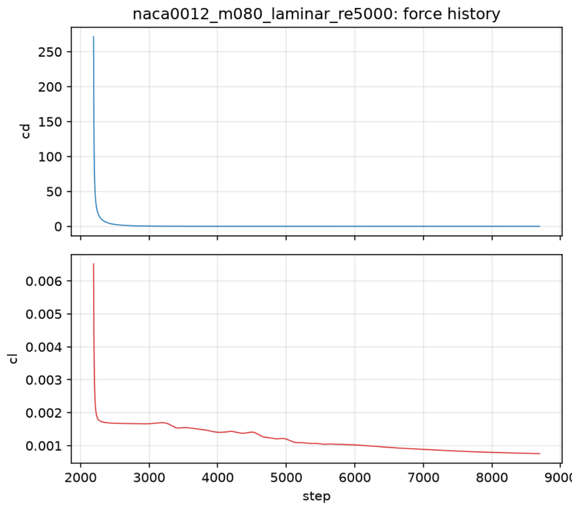


(e) Static-pressure field from `field_final.vtu`.

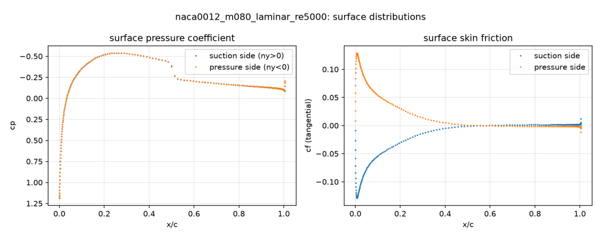
Figure 4: NACA0012, $M_\infty = 0.15$, laminar, $Re = 5000$, $\alpha = 0$.



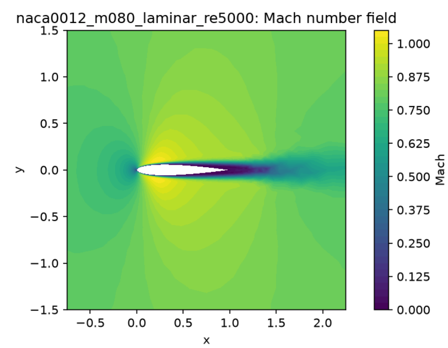
(a) Residual history from `residuals.csv`.



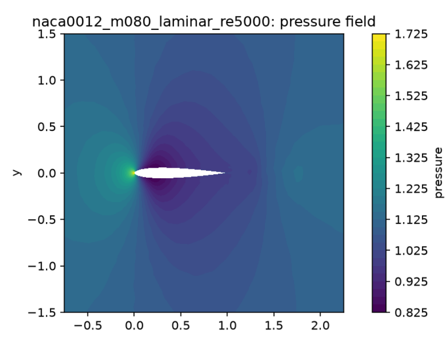
(b) Force history from `forces.csv`.



(c) Surface C_p and C_f from `surface.csv`.

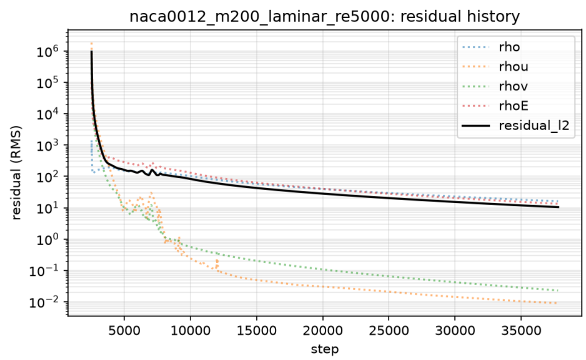


(d) Mach-number field from `field_final.vtu`.

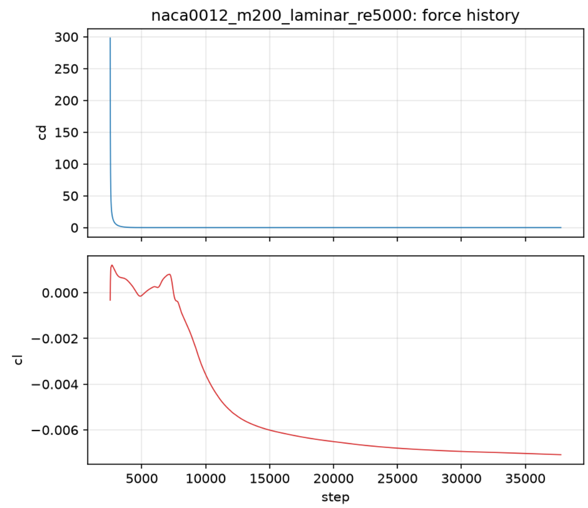


(e) Static-pressure field from `field_final.vtu`.

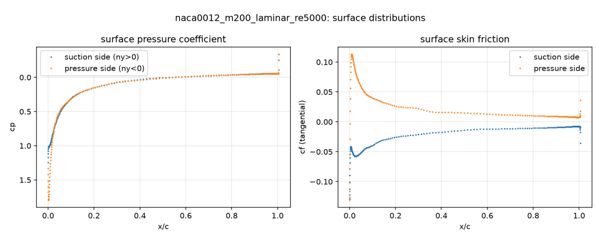
Figure 5: NACA0012, $M_\infty = 0.8$, laminar, $Re = 5000$, $\alpha = 0$.



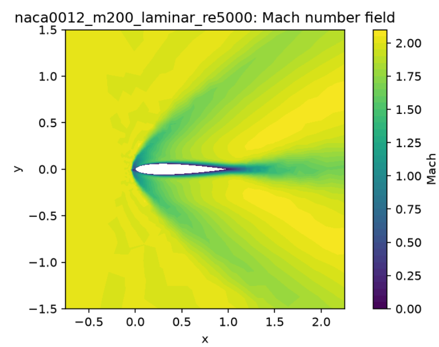
(a) Residual history from `residuals.csv`.



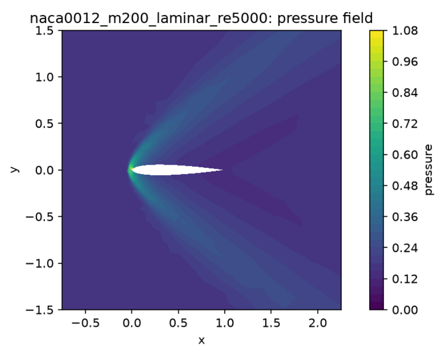
(b) Force history from `forces.csv`.



(c) Surface C_p and C_f from `surface.csv`.

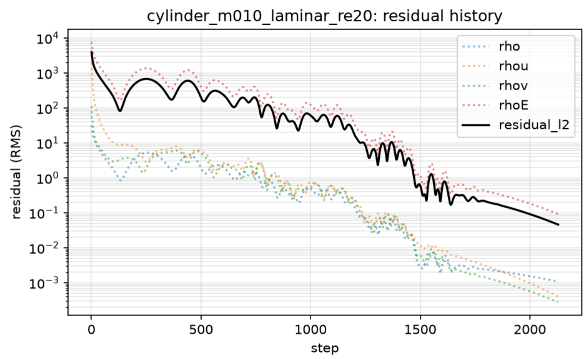


(d) Mach-number field from `field_final.vtu`.

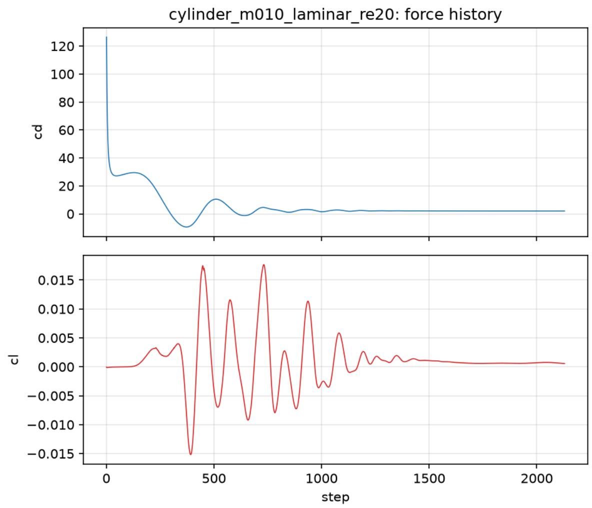


(e) Static-pressure field from `field_final.vtu`.

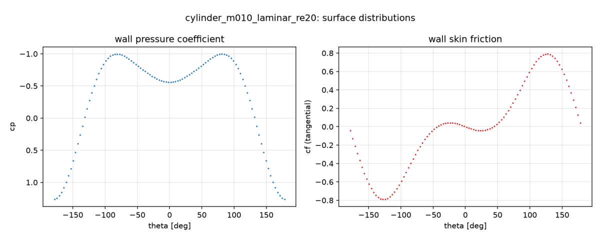
Figure 6: NACA0012, $M_\infty = 2.0$, laminar, $Re = 5000$, $\alpha = 0$ (production redo with CFL cap 5 and 5-order residual target).



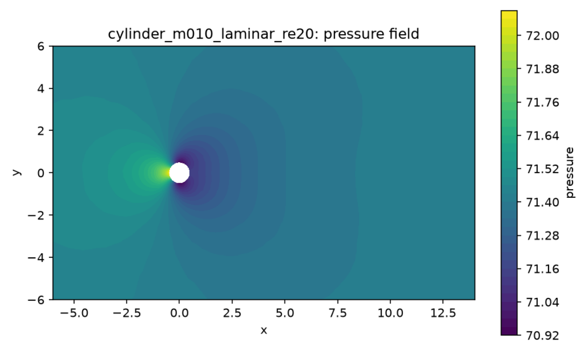
(a) Residual history from `residuals.csv`.



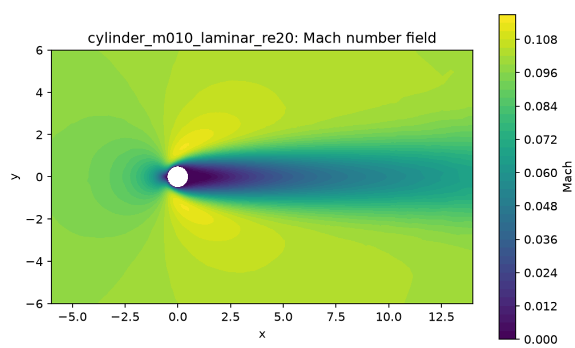
(b) Force history from `forces.csv`.



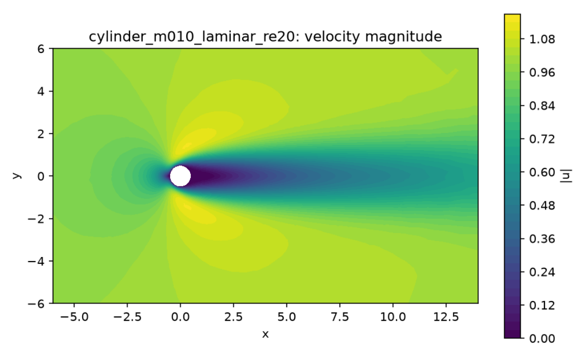
(c) Surface C_p and C_f from `surface.csv`.



(d) Static-pressure field from `field_final.vtu`.



(e) Mach-number field from `field_final.vtu`.



(f) Velocity-magnitude field (wake zoom) from `field_final.vtu`.

Figure 7: Cylinder, $M_\infty = 0.1$, laminar, $Re = 20$ (steady).

1.020 + viscous 0.242), the mean lift is $C_L = -9.9 \times 10^{-4}$, the lift oscillates with half-range ≈ 0.57 (peak-to-peak ≈ 1.13; the windowed-FFT amplitude returned by `spectral_stats` is 0.55), and the drag oscillation has FFT amplitude 0.024 at twice the lift frequency ($f = 0.373$) — the expected drag pulse twice per shedding cycle. The mean moment coefficient is $C_{m_z} = 3.0 \times 10^{-5}$. These values are consistent with the literature for laminar vortex shedding at $Re = 200$ ($St \approx 0.19$–0.21, mean $C_D \approx 1.3$–1.4); the wake vorticity (fig. 8c, clipped to $[-5, 5]$ per the case-file recommendation) and the pressure and velocity fields (figs. 8d and 8e) show the developed Kármán street.

10 Parallel Validation

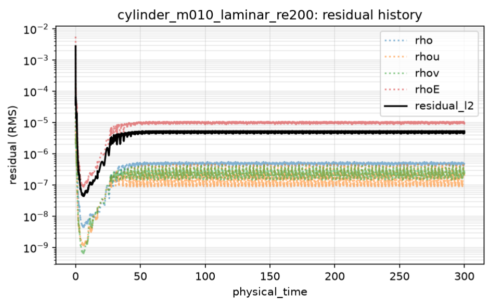
Two cases were run across $np \in \{1, 2, 4, 8\}$ in a dedicated back-to-back sweep (`tools/rank_sweep.sh`, outputs under `results/rankstudy/`); table 6 collects the results. Every rank count re-partitions the mesh (edge cut rises with np: 122 at $np = 2$ to 422 at $np = 8$ for the airfoil), so the SGS sweep order and the approximate Jacobian differ slightly between rank counts; with the halo-consistent inner solve the step counts to reach the 4–5 order targets are nevertheless identical across rank counts (1482 steps for the airfoil case, 2129 for the cylinder case), and the converged states agree as follows.

Table 6: Rank study: final forces, steps, and wall times across MPI rank counts for `naca0012_m015_inviscid` and `cylinder_m010_laminar_re20`, all from one back-to-back sweep on the shared host (load average 28–46 during the window). Wall times at these small mesh sizes are indicative rather than HPC-grade; force consistency and run-to-run determinism are the validation metrics.

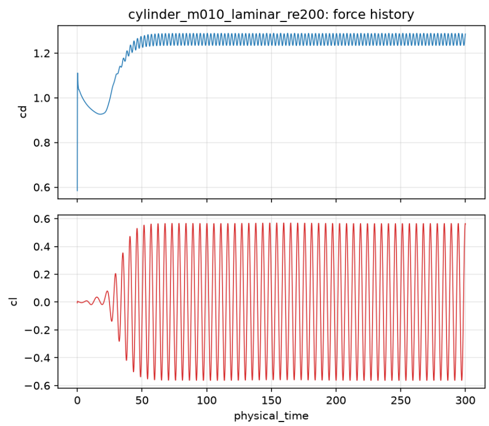
Case	np	Steps	Orders	C_D	C_L	Wall [s]
`naca0012_m015_inviscid`	1	1482	4.03	0.001705	−1.94e−4	51.4
	2	1482	4.03	0.001702	−1.88e−4	27.3
	4	1482	4.02	0.001693	−1.88e−4	15.9
	8	1482	4.02	0.001696	−2.00e−4	22.6
`cylinder_m010_laminar_re20`	1	2129	5.00	2.01895	5.63e−4	33.6
	2	2129	5.00	2.01896	5.58e−4	17.7
	4	2129	5.00	2.01895	5.51e−4	9.9
	8	2129	5.00	2.01896	5.63e−4	16.6

Force consistency. For the cylinder case the final drag agrees across rank counts to within 4×10^{-6} in C_D (relative 2×10^{-6}) and lift to $\pm 3 \times 10^{-5}$ — the partitioning changes only the convergence path, not the converged state. For the inviscid airfoil case the absolute drag is itself only $\mathcal{O}(10^{-3})$ (spurious inviscid drag, section 9.2); the rank-to-rank scatter (1.693×10^{-3} to 1.705×10^{-3}) is below one percent of that floor, and the lift scatter is $\pm 6 \times 10^{-6}$ about the mean. There is no systematic rank-dependent drift in either case, consistent with the global-id-ordered, rank-count-independent restart files and the no-replication assertions of section 7. Moreover, the $np = 8$ sweep runs reproduce the production $np = 8$ runs exactly — identical step counts (1482 and 2129) and identical final forces to all printed digits — demonstrating run-to-run determinism at fixed rank count and partition.

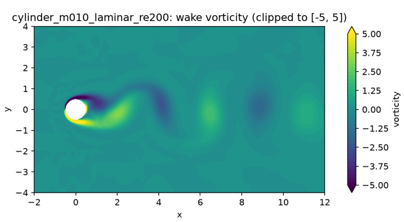
Timing and load balance. Within the sweep (one machine-load window), wall time scales from 51.4 s ($np = 1$) to 15.9 s ($np = 4$) for the airfoil — a 3.2× speedup, 81% parallel efficiency — and



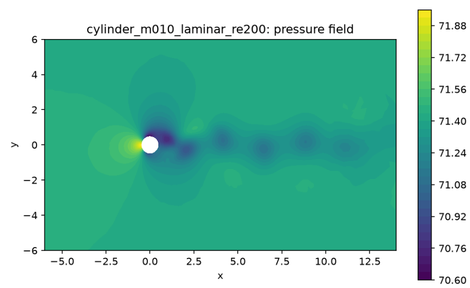
(a) Inner-loop residual ratio from `residuals.csv`.



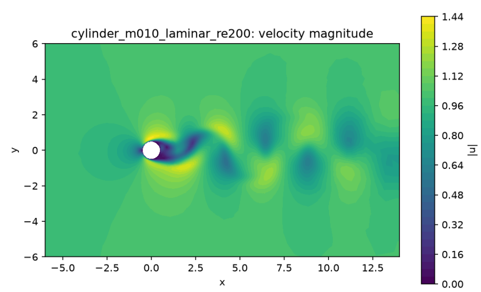
(b) Force history (C_D, C_L) from `forces.csv`.



(c) Wake vorticity ω_z, clipped to $[-5, 5]$, from the VTU snapshots.



(d) Static-pressure field from the VTU snapshots.



(e) Velocity-magnitude field from the VTU snapshots.

Figure 8: Cylinder, $M_\infty = 0.1$, laminar, $Re = 200$, BDF2 $\Delta t = 0.01$, statistically periodic state at $t = 300$.

from 33.6 s to 9.9 s for the cylinder (3.4×, 85%). At $np = 8$ both cases run slightly slower than at $np = 4$ (22.6 s and 16.6 s): with only ≈2600 (airfoil) or 1300 (cylinder) owned cells per rank, the halo exchange and per-step `MPI_Allreduce` latency are comparable to the compute, and the shared host adds noise at these runtimes. Load balance from the partition diagnostics is 1.03 or better at $np = 8$ (table 3); communication volume remains small (edge cut $\leq 2.4\%$ of faces). For reference, the production $np = 8$ runs recorded 254.7 s (airfoil) and 88.1 s (cylinder) while sharing the host with three other production runs — run-to-run timing variability on this shared host is exactly why the sweep timings above were all taken back-to-back.

11 Visualization and Plot Style Requirements

All figures are generated programmatically by `tools/plot_results.py` from the run artifacts and are listed with their source files in `report/figure_manifest.csv`; no screenshots are used.

1. Line plots (residual histories, force histories, surface distributions) use readable font sizes, 1.5–2 pt line widths, labeled nondimensional axes (pseudo step versus $\log_{10}$ residual RMS; physical time versus C_D, C_L; surface angle versus C_p, C_f), legends, and grid lines, with consistent colors per quantity across cases.

2. Field plots are triangulated filled contours (cell values averaged to nodes) with labeled colorbars naming the plotted variable, near-body zoom windows that resolve the airfoil boundary layer / shock and the cylinder wake, and clipped ranges where an automatic range would hide the structure (wake vorticity clipped to $[-5, 5]$).

3. Both Mach and pressure contours are provided for every case, and file names match the plotted variable exactly (`field_mach_<case>.png` plots Mach, `field_pressure_<case>.png` plots pressure, `field_velmag_<case>.png` plots velocity magnitude, `wake_vorticity_<case>.png` plots ω_z).

4. Per-case figures are laid out as compact subfigure grids (this report uses `subcaption` blocks) rather than sparse one-plot pages, and every figure is introduced in the text with a cross-reference.

12 Code Organization and Extensibility

The solver is split into single-responsibility modules (`src/`): `mesh.cpp` (CGNS import, cell/face geometry), `partition.cpp` (METIS graph partitioning, ghost layer, halo plan, per-rank partition cache), `spatial.cpp` (least-squares gradients, limiters, inviscid/viscous flux assembly, boundary conditions), `timestep.cpp` (steady and BDF2 transient drivers, LU-SGS inner solves, force reduction), `output.cpp` (CSV/VTU/restart/diagnostic writers), `case_file.cpp` (JSON parsing and reference values), and `physics.hpp` (conserved/primitive conversion, Roe and Rusanov fluxes, exact flux Jacobian, viscous stresses). No supplied case is hard-coded anywhere in solver logic: boundary-family-to-condition mapping, freestream state, gas model (γ, R, Pr), reference quantities, and run controls are all parsed from the case JSON, and case-to-case variation enters only through input files or documented command-line options. Extension space is deliberate: a general equation of state replaces the calorically-perfect relations in `physics.hpp` behind the `GasModel` interface; RANS adds an eddy-viscosity field consumed by `viscous_flux_phys` plus transport/source terms in the residual driver; multi-species and 3-D extend the state vector and the geometry/face loops, which are written in terms of face normals and cell volumes rather than 2-D-specific formulas.

13 Limitations and Failure Analysis

Spurious inviscid drag floor. In inviscid subsonic flow at zero angle of attack the physical drag vanishes (d'Alembert), yet the $M0.15$ inviscid case converges with $C_D = 0.0017$: limiter dissipation near the stagnation region and mild grid skewness break the exact fore-aft cancellation. The value sets the drag floor of this mesh/scheme combination for smooth inviscid flow; the transonic and supersonic inviscid wave drags are well above it. An earlier solver build stalled below the $M2.0$ inviscid residual target (a missing halo exchange of the inner-solve increment across partition boundaries, since fixed); the shipped solver converges to all supplied inviscid targets directly.

High-Mach laminar escalation. The $M0.8$ and $M2.0$ laminar cases could not be run cold at their supplied CFL schedules: the shock/boundary-layer stiffness at high pseudo-CFL produced a higher-CFL instability. Both were therefore restarted from their converged inviscid states with the CFL capped (4 and 5 respectively, versus supplied maxima of 80 and 50). The $M0.8$ laminar restart then converged cleanly to 4.00 orders at $C_D = 0.0701$.

$M2.0$ **laminar: under-converged first run, divergence, redo.** The first $M2.0$ laminar run used the case-file 3-order residual target and appeared converged while the boundary layer was still developing from the inviscid restart: the restart transient falls 3 orders almost entirely during the re-adjustment of the shock system (at -2.8 orders from the restart baseline the drag was still $C_D \approx 4.5$, and it fell through 1.2 by -3.5 orders), so a 3-order stop lands mid-development at $C_D \approx 3.2$, an order of magnitude above the plausible window. That run would have been misreported as successful had the force level not been checked. A debug continuation (`results/dbg_m200lam_continue/`) then diverged once the CFL ramp exceeded ≈ 7 (non-finite residual at step 47933, marked `failed`). The production redo used `--cfl-max 5` and `--residual-target 5.0` and converged to 5.00 orders in 37786 total steps (35258 after the restart), 1229 s, with $C_D = 0.1315$ stable to 10^{-4} over the last 500 steps. The lesson — residual-orders alone are not a safe convergence criterion for restarted viscous cases — is why the sanity checks of section 8 include a force-plausibility window.

$Re = 200$ **statistics window.** The $Re = 200$ vortex-shedding run completed to $t = 300$ and is marked `statistically_periodic` by the solver's own late-window periodicity test, and the post-transient statistics ($St = 0.187$, mean $C_D = 1.26$) sit inside the literature ranges. The statistics are nevertheless computed on a 150-time-unit window, so the FFT frequency resolution is coarse ($\Delta f \approx 1/150$) and the quoted amplitudes carry a few-percent window-to-window variability; a longer integration would tighten them.

Further improvements with more time. (i) A shock-robust or frozen-limiter variant to reduce the spurious inviscid drag floor further; (ii) mesh refinement in the airfoil boundary layer and cylinder wake to reduce the spurious inviscid drag floor and tighten the $Re = 200$ statistics (a longer time window would also tighten the FFT frequency resolution, currently $\Delta f \approx 1/150$); (iii) replacing the simplified Rusanov Jacobian with a Roe-consistent Jacobian in the inner solve to cut inner iterations at high Mach.